Research Working Group

bioinfo 4women

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3. *Medicines Discovery Catapult*
4. *DEVCENTER - Advanced Technology*

INTRODUCTION

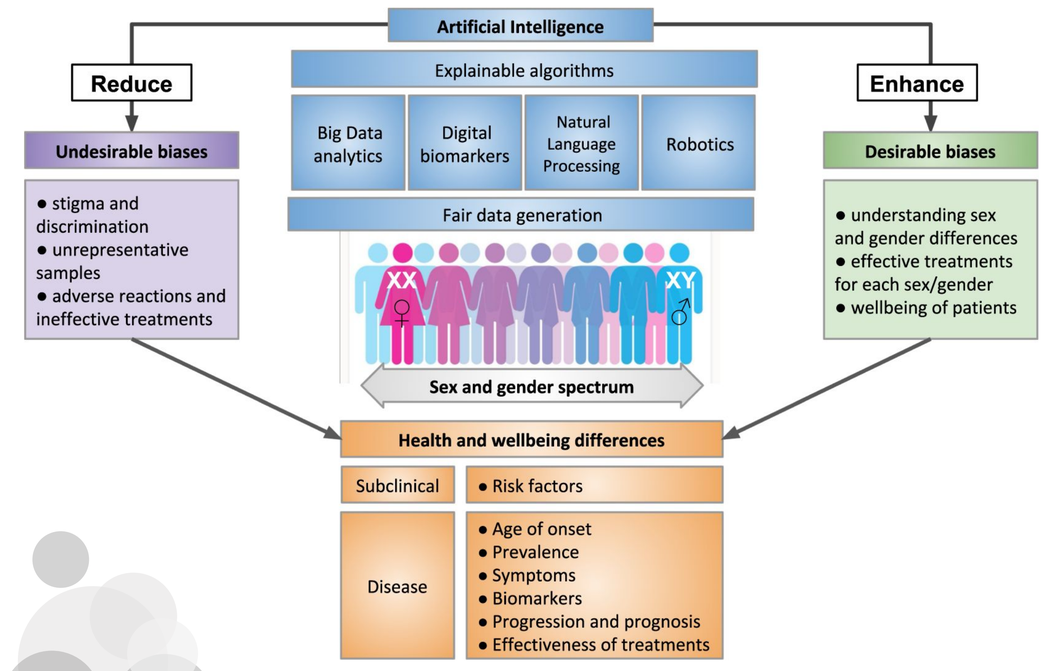
OPPORTUNITIES & CHALLENGES

Artificial Intelligence has a huge **impact on our society**, in particular biomedical research and **decision making in clinical research**.

Cost-efficiency, dynamic access to health information, effective handling of complex problems related to human diseases.

Identify unnoticed bias towards sex and gender categories could potentially lead to propagation of **discrimination and inequality**.

Understand sex and gender differences medically relevant and needed in training dataset to achieve **fair Precision Medicine**



Cirillo D., Catuara-Solarz, S. et al. npj Digit. Med. 3, 81 (2020) DOI: 10.1038/s41746-020-0288-5

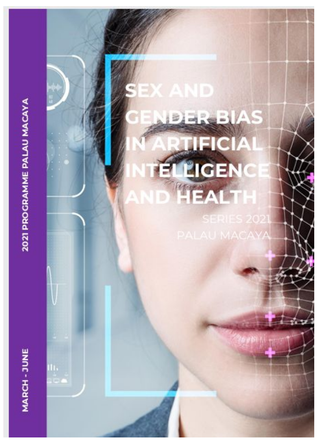
RAISING AWARENESS

Objectives

1. Provide a space for **open debate** between scientific community and society on sex and gender biases in science and technology.
2. Identify **strategies to disseminate scientific results and educate society.**
3. Contribute to the achievement of UN's **Sustainable Development Goals.**

Actions

- Sex and Gender Bias in Artificial Intelligence and Healthcare project, funded by **La Caixa Foundation at Palau Macaya** between March and June 2021.
- 4 conferences, 2 seminars with round tables, 2 workshops.
- National and international experts with multidisciplinary background.
- Publication of final report and working on a scientific paper.



***Reference**:*
Sustainable Development Goals. United Nations. https://sdgs.un.org/goals

BUILDING BEST PRACTICES

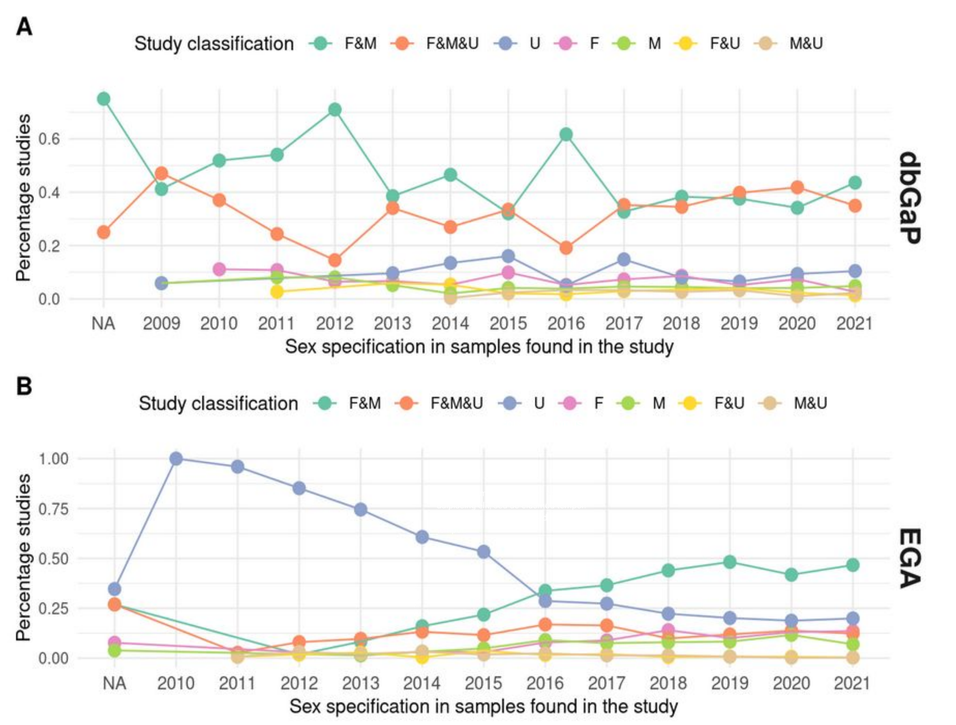
Objectives

1. **Identify sex and gender biases** in data acquisition processes.
2. **Create a repository** of public biomedical databases and scientific publications.
3. Elaborate **guidelines of best practices** for research in Artificial Intelligence and Precision Medicine.

Actions

- FAIR-X project, **ELIXIR Biohackathon 2021**
- **Quantitative analysis to measure biases** in large-scale public biomedical resources using dbGaP and EGA databases
- **Qualitative analysis on data collection and processing**
- Propose a **set of recommendations** based on the results obtained

*Quantification and sex classes distribution of **(A)** dbGaP and **(B)** EGA samples. Both panels show studies classification according to the sex representation of their samples across time. F, female; M, male and U, unknown. 'NA' refers to no date record.*



CONCLUSIONS

Clear definitions

Sex as biological characteristic; **Gender** as sociocultural behaviour, attitude and identity

- **Avoid confusion** in data collection & processing.

Standarization

Improve **data management** and quality

- Enable **interoperability**, **reutilisation** & **open access.**

Social impact

Analyse, understand, disseminate the effects of precision medicine in underrepresented communities.

Stakeholder participation

Include **participatory methods and multidisciplinarity** in design, development, validation and use phases.

Privacy-preserving

Scientific research aligned with **ethical and normative framework**

- GDPR, anonymisation techniques, synthetic data, genomic data.

Education

Raise awareness & fix the knowledge

- Include diversity in dataset creation, research practice and university curricula.

Transparency

Foster **open science and access**, reusability. Create **user-friendly quality indicators** for dataset creators and curators

- Enhance **accountability**